

Course Syllabus

Welcome to CS 3240

Discrete Mathematical Structures II (Theory of Computation)

Software developers are in the business of using computers to solve real problems for real people. They need to know what computers are capable of, and, just as important, they need to know **what computers can not do**. To fully understand the effective use of computers as a problem-solving tool, it is imperative to have a thorough grasp on the **foundations of computation**. In this class we examine the theoretical underpinnings of automatic computation and their modern applications.

This course gives you an understanding of the theoretical foundations of automatic computation. You will understand the **basic models** and the **inherent limitations of machine computation**. It is interesting that most of our material was developed before computers were invented, and in fact had some influence on how computers came to be.

Student Learning Outcomes

- Understand the concepts of **Regular Languages** and their relationship to **finite state machines** (automata)
- Understand the concepts of **Context-Free Languages** and their relationship to **pushdown automata**
- Understand the concepts of **Computable Languages** and their relationship to **Turing machines**
- Apply all of the above in computing
- Understand the limits of automatic computation

Prerequisites

CS 2300 (Discrete Structures I), **CS 2420** (Data Structures), **CS 2810** (Computer Architecture and Assembly Language)

Text

Allison, [*Foundations of Computing*](#), 2021. You are provided a **free PDF** (click on the Files folder). If you wish to have a paperback book version, you may purchase one on amazon.com.

Note: This text has been created specifically for this class. We will use almost all of it. Read the assigned sections thoroughly and do the associated reading exercises *before class* and you stand a better chance of mastering the material. You will be tested on assigned material *in the book* as well as on what we cover in class.

Grading

Your performance on the following will determine your course grade:

Reading **Exercises** 5%

Homework 45%

2 Mid-term **Exams** 30% (15% each)

Final Exam	20%
Pop Quizzes	5% (extra credit; in-class only)

The Reading Quizzes are short and are intended to be done while you are reading. You have 3 attempts, so do them right away as you read—you can repeat if you don't get all the points at first.

The quiz extra credit bonus is contingent on your completing the online course evaluation (SRI). Submit an electronic copy of your SRI receipt in Canvas.

Course Schedule

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Week	Date	Assignments
Week 1	Jan 10	Chapter 1, Section 2.1
Week 2	17	Sections 2.2
Week 3	24	Section 2.3–2.4
Week 4	31	Chapter 3
Week 5	Feb 7	Sections 4.1–4.2
Week 6	14	Section 4.3
Week 7	21	Chapter 5
Week 8	28	Sections 6.1–6.2
Week 9	Mar 7	Spring Break
Week 10	14	Section 6.3
Week 11	21	Sections 7.1–7.2
Week 12	28	Sections 7.3–7.4
Week 13	Apr 4	Chapter 8
Week 14	11	Chapter 8
Week 15	18	Chapter 9
Week 16	25	Chapter 10